Metrology
(Weight and volume measures, dose calculations)

Pharmacology Lab

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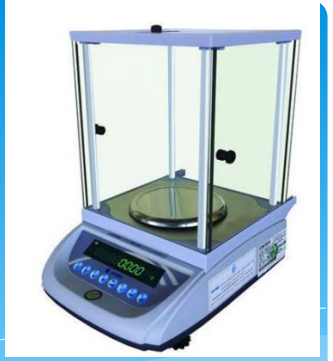
Introduction

* Metrology: is the study of the various systems of weights and measures and their relationships and the mathematical principles involved in their inter conversion.

* It Is very important since any mistake in calculations or measurements could lead to either under dosing or over dosing. This in turn leads to, inadequate treatment or toxicity.

Measurement of weight



* Weight is measured by means of a balance. There are different types of balance, types for rough weights and others for sensitive weights.

Rules for the use and care of the balance

1-the balance should be located in a well –lightened place ,as far as possible from vibration ,dust, moisture and corrosive vapor.

2-the balance should be kept clean at all times.

3-watch glass should be used for weighing of solids, stoppered bottles used for corrosive substances and liquids

4-check to see if the balance is adjusted including zeroing of it ,also check the adjustment before each weighing.

Using an Electronic Balance

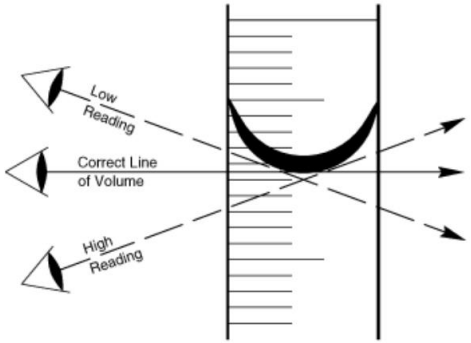
Measurement of volume

* Factors considering liquid measurements:

1- The choice of proper measuring vessel (Do not measure small volume by large graduates).

2- The surface area of the liquid in the measuring vessel. The accuracy increases as the surface area decreases.

3- Density and surface tension of the liquid should be considered.



❖ Techniques of measuring liquid:

1- Pour the liquid from the bottle to the graduate cylinder.

2- The graduate cylinder is then raised to level of eyes to minimize the error in reading.

3- The liquid is poured into a graduate cylinder until the bottom of the meniscus exactly reaches the required mark

Wise-Online.com

Systems of measurement

* Three types of systems for weight and volume are used:

1. Old apothecaries' system
2. The classical metric system
3. Household (domestic) measurement

1. Old apothecaries' system

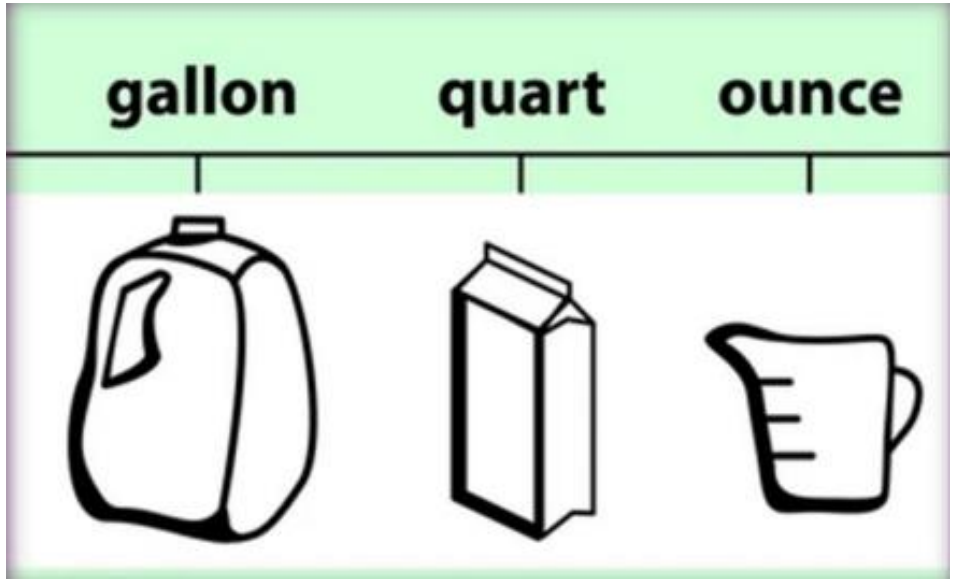
* Sometimes referred to as “Imperial” or “British-American”, “English” or “British Imperial”
* Not used much but some of these units are still in use especially in UK and in US.

Examples:

Volume: fluid ounce (fl.oz), pint (pt), quart (qt.) and gallon (gal.)

Weight: grain (gr.), ounce (oz.), pound (lb.) and stone (St.)

unit	type	examples	
1 ounce	weight	5 quarters, candy bar, cheese slice	
1 pound	weight	2 apples, box of cereal, pair of sneakers	
1 ton	weight	small car, 5 large bears	
1 fluid ounce	volume	bottle of perfume, tube of sunscreen	
1 cup	volume	coffee cup, juice glass, jar of honey	
1 pint	volume	bottle of water, small ice cream container	
1 quart	volume	quart of milk, can of motor oil, medium ice cream container	
1 gallon	volume	gasoline, large milk jug, two large ice cream containers	



2. The classical metric system

* International System of Units (SI)
* An internationally recognized decimal system, formerly known as “metric system”; formulated in France in the late 18th century.
* For liquid dosage forms, the concentration of the drug can be represented per dose, per container contents, per unit measurement or simply as a percentage.

Examples:

Volume: milliliter (ml or cc cubic centimeter), Liter (L)

Weight: microgram (mcg), milligram (mg), gram (g), kilogram (kg)

3. Household (domestic) measurement

* They are not accurate enough to be used in the calculation of drug dosage in hospitals. However, it is easy for the patient to understand and deal with it. Therefore, its used by physician to prescribe the dose to their patients.

Examples:

Teaspoon (tsp), Tablespoon (Tbsp),

coffee cupful (ounce), cup (glassful).

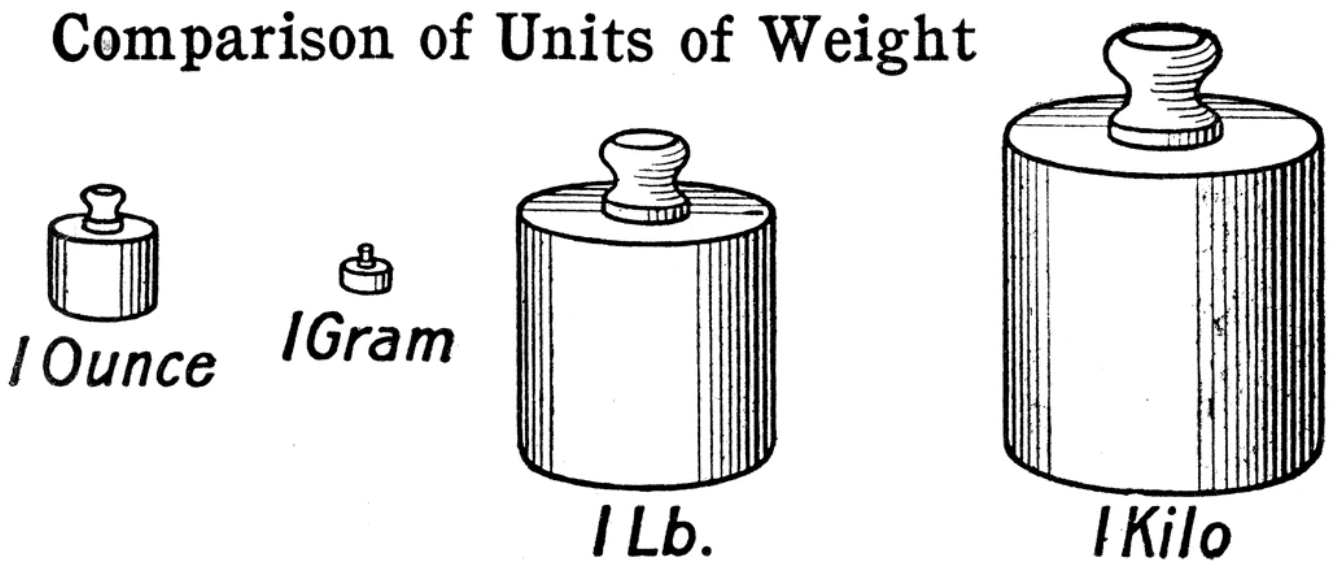
What Does A Tablespoon Look Like

Most common conversions between systems

* To convert weight units from apothecary to metric system and vice versa we use the following:

1 oz. = 30 g

1 kg = 2.2 lb.



Most common conversions between systems

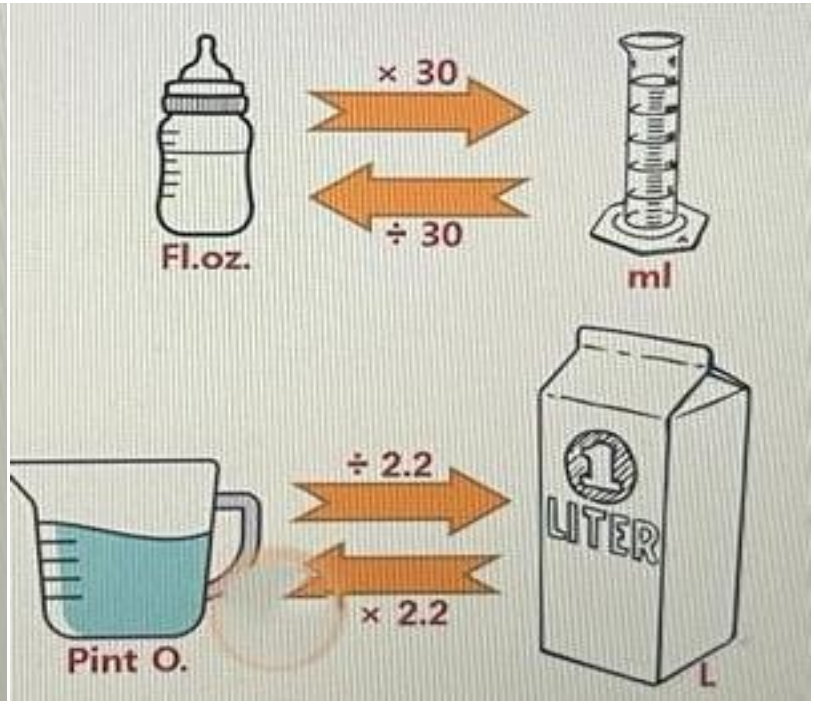
* To convert units of measuring liquids from apothecary to metric system and vice versa we use the following:

1 fl.oz. = 30 ml

1 L= 2.2 O

1 O = 450 ml

1ml = 20 drops
1 tsp = 5ml



Dose Calculations

* POSOLOGY : is the study of dose quantity of a drug.
* Quantitatively, drug doses vary greatly among drug substances; some drugs have small doses, other drugs have relatively large doses
* The dose of a drug is based on its biochemical land pharmacologic activity, its physical and chemical properties, the dosage form used, the route of administration, and various patient factors.
* The adult dose of a drug is the amount that ordinarily produces the medicinal effect intended in the adult patient.
* The pediatric dose is similarly defined for the infant or child patient.
* Calculation of adult dose quantity for the children based on their age , weight or body surface area

Reasons for calculating pediatric dose

* Pediatrics is the branch of medicine that deals with disease in children from birth through adolescence. Because of the range in age and bodily development in this patient population, the inclusive groups are defined further as follows: neonate (newborn), from birth to 1 month; infant, 1 month to 1 year; early childhood, 1 year through 5 years; late childhood, 6 years through 12 years; and adolescence, 13 years through 17 years of age
* Factors affecting on pediatric patients in drug dosing

- patients age
- weight
- over all health state

➢ The condition of such biologic function as respiration and circulation and the stage of developments of body system for drugs metabolism (e.g., liver enzymes) and drug elimination (e.g., renal system).

* • In neonates the kidney function developed over the span of the first 2 years of life so we have to be very careful when the anti microbial and other drugs which are determined by the kidney since This fact is particularly important because the most commonly used drugs in neonates, infants, and young children are antimicrobial agents which are eliminated primarily through the kidneys. If the rate of drug elimination is not properly considered, drug accumulation in the body could occur, leading to drug overdosage and toxicity

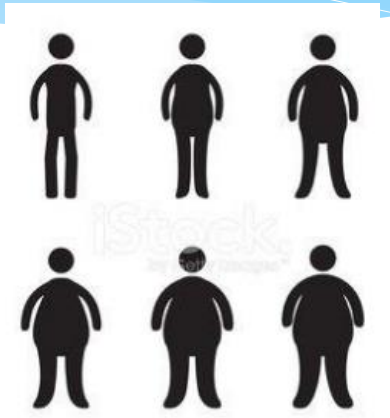
Geriatric dose

* Although the term elderly is subject to varying definitions with regard to chronologic age, it is clear that the functional capacities of most organ systems decline throughout adulthood, and important changes in drug response occur with advancing age. Geriatric medicine the field that encompasses the management of illness in the elderly.\
* **Special Considerations in Dose Determinations for Elderly Patients** (Dose determinations for elderly patients frequently require consideration of some or all of the following):

1. Therapy is often initiated with a lower-than-usual adult dose.
2. Dose adjustment may be required based on the therapeutic response.
3. The patient's physical condition may determine the drug dose and the route of administration employed.
4. The dose may be determined, in part, on the patient's weight, body surface area, health and disease status, and pharmacokinetic factors.
5. Concomitant drug therapy may affect drug/dose effectiveness.
6. A drug's dose may produce undesired adverse effects and may affect patient compliance.
7. Complex dosage regimens of multiple drug therapy may affect patient compliance.

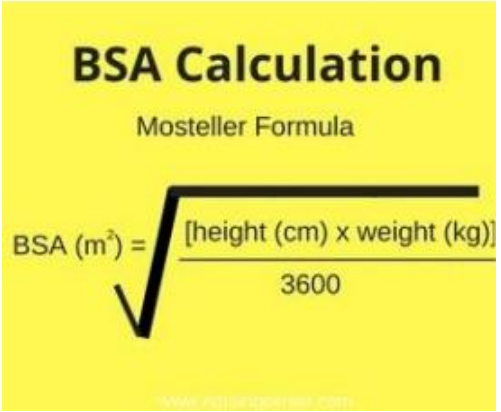
Classification of dose

* BASED ON AGE GROUP



* BASED ON WEIGHT

* BASED ON BODY SURFACE AREA(BSA)



AGE GROUP

DOSE BASED ON AGE GROUP

a)Young's rule

* For children having age between 1years – 12 years

$$\text{Dose quantity} = \frac{\text{Age of child in years}}{\text{Age of child} + 12} \times \text{adult's dose}$$

Example:

Age=10 years

Adult's dose = 150mg

Sol:

$$\frac{10}{10 + 12} \times 150 = 68.2 \text{ mg}$$

DOSE BASED ON AGE GROUP

b) Fried's rule

* It is use to calculate dose for INFANTS (new born baby) 1-12 months.

$$\text{Dose Quantity} : \frac{\text{Age (month)}}{150} \times \text{Adult dose}$$

Example:

Age in months = 10 months

Adult's dose = 200mg

Sol:

$$\frac{10}{150} \times 200 = 13.33$$

The value of 150 in Fried's rule was an estimate of the age (12.5 years or 150 months) of an individual who would normally receive an adult dose,

DOSE BASED ON AGE GROUP

c) Dilling's rule

* For the girl having age from 12years- 18years.

$$\text{Dose quantity} = \frac{\text{age in years}}{20} \times \text{adult's dose}$$

Example:

Age of girl = 15

Adult's dose = 200mg

Sol:

$$\frac{15}{20} \times 200 = 150$$

DOSE BASED ON AGE GROUP

d) Cowling's rule

* If the age of child is given in years then age at next birthday should be taken.

$$\text{Child Dose} : \frac{\text{Age of child in years} + 1}{24} \times \text{Adult dose}$$

DOSE BASED ON WEIGHT

CLARK'S RULE

* kg or Pounds (lbs)
* 1kg = 2.2 lbs

$$\text{Dose quantity} = \frac{\text{weight in (kg) OR (Lbs / pound)}}{\text{68.8 kg OR 150 Lbs}} \times \text{adult's dose}$$

In some cases, the usual dose is expressed as a specific quantity of drug per unit of patient weight, such as milligrams of drug per kilogram of body weight (abbreviated mg/kg). Dosing in this manner makes the quantity of drug administered specific to the weight of the patient being treated.
The patient's weight is an important factor in dosing since the size of the body influences the drug's concentration in the body fluids and at its site of action.
Dose calculations based on body weight have become standard for certain drugs in dosing both adult and pediatric patients.

The number 150 in Clark's rule was an estimate of the weight of an individual who likewise would receive an adult dose.

DOSE BASED ON BODY SURFACE AREA (height and weight)

* It can be used for two types of patients

A) Cancer patients receiving chemotherapy.

B) Pediatric patients.

BSA Formula

$$\frac{\text{Body surface area of child}}{\text{Body surface area of adult}} \times \text{Adult dose} = \text{Child's dose (approx)}$$

The average body surface area for an adult has been given as $1.73\ m^2$; hence,

$$\frac{\text{Body surface area of child } (m^2)}{1.73\ m^2} \times \text{Adult dose} = \text{Child's dose (approx)}$$

If the adult dose of a drug is 100 mg, calculate the approximate dose for a child with a BSA of
0.83 m2

$$\text{Child's dose} = \frac{0.83\ m^2}{1.73\ m^2} \times 100\ mg = 47.97 \text{ or } 48\ mg$$

General Notes

* Most drugs in children are dosed according to body weight (mg/kg) or body surface area (BSA) (mg/m2). Care must be taken to properly convert body weight from pounds to kilograms (1 kg= 2.2 lb.) before calculating doses based on body weight. Doses are often expressed as mg/kg/day or mg/kg/dose, therefore orders written "mg/kg/d," which is confusing, require further clarification from the prescriber.

* Chemotherapeutic drugs are commonly dosed according to body surface area, which requires an extra verification step (BSA calculation) prior to dosing. Medications are available in multiple concentrations, therefore orders written in "mL" rather than "mg" are not acceptable and require further clarification.

* Dosing also varies by indication, therefore diagnostic information is helpful when calculating doses. The following examples are typically encountered when dosing medication in children.

Example 1

* Four months infant weighing 17.6 lb. with fever. Calculate the dose of paracetamol in (ml) for this patient, and prescribe it for his mother. The dose of paracetamol is 10mg/kg/dose given every 4-6 hours, and it comes in concentration of 100mg/ml
* Answer

1st convert lb. to Kg ⟶ $17.6 \div 2.2 = 8\ kg$

2nd calculate the dose in ml ⟶

10 mg 1 kg

? 8kg

80 mg/ dose (every 4-6 hours)

the suspension comes in a concentration of 100 mg/ml ⟶ 100 mg in each one ml of suspension, so the dose in (ml) calculated

100mg 1 ml

80 mg ?ml

0.8 ml

❖ 3rd prescribing the dose to the mother

If the mother cannot understand how she can give 0.8 ml of paracetamol suspension to her baby. We will simplify it by

There are approximately 20 drops in 1 milliliter (ml)

20 drops 1ml

? 0.8 ml 16 drops

Example 2

Q/ Calculate the dose of amoxicillin suspension in mls for otitis media for a 1-yr-old child weighing 22 lb. The dose required is 40 mg/kg/day divided BID and the suspension comes in a concentration of 400 mg/5 ml.

Step 1. Convert pounds to kg:	22 lb × 1 kg/2.2 lb = 10 kg
Step 2. Calculate the dose in mg:	10 kg × 40 mg/kg/day = 400 mg/day
Step 3. Divide the dose by the frequency:	400 mg/day ÷ 2 (BID) = 200 mg/dose BID
Step 4. Convert the mg dose to mL:	200 mg/dose ÷ 400 mg/5 mL = **2.5 mL BID**

Example 3

An over-the-counter cough remedy contains 120 mg of dextromethorphan in a 60-mL bottle of product.
The label states the dose as 1½ teaspoonfuls for a child 6 years of age. How many milligrams of dextromethorphan are contained in the child's dose?

$$1\tfrac{1}{2}\text{ teaspoonfuls} = 7.5\text{ mL}$$

$$\frac{60\text{ mL}}{120\text{ mg}} = \frac{7.5\text{ ml}}{x\text{ mg}}$$

$$x = 15\text{ mg dextromethorphan,}$$

THANK YOU :)